Context-Aware Optimization for Autonomous Agents

A Protocol-Driven Framework for Multi-Objective Policy Search

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Abstract We present CIE (Context-aware Introspection and Evaluation), a terminal-based framework for optimizing autonomous agent policies through multi-objective evaluation and hierarchical context management. Unlike traditional optimization approaches that treat agent context as opaque state, CIE introduces explicit context introspection—allowing agents to inspect, compress, and reorganize their working memory during policy search. Our system combines protocol-driven extensibility with Pareto-optimal multi-objective optimization, enabling practitioners to balance competing metrics (latency, cost, success rate) while maintaining operational guardrails. Through a comprehensive benchmark suite spanning navigation, comprehension, and modification tasks, we demonstrate that context-aware optimization yields 2.3× compression ratios and 40% improvements in access pattern efficiency. The framework supports multiple optimization algorithms (DSPy few-shot learning, adaptive hill climbing) and provides both interactive TUI and CLI interfaces for research and production deployment. Our approach bridges the gap between prompt engineering workflows and systematic policy optimization, offering a path toward more interpretable and efficient autonomous systems.

Contents

1 Introduction

The optimization of autonomous agent policies remains a central challenge in artificial intelligence research. While recent advances in large language models have enabled sophisticated reasoning capabilities, the process of refining agent behavior—selecting optimal prompts, balancing computational resources, and maintaining reliability—continues to rely on manual iteration and heuristic tuning (Brown et al., 2020; Wei et al., 2022). This gap between model capability and deployment efficiency motivates the need for systematic optimization frameworks.

Consider a typical agent development workflow: practitioners experiment with different prompt templates, adjust model parameters, and evaluate performance across diverse workloads. Each modification requires re-running evaluations, comparing metrics, and deciding whether changes represent genuine improvements or noise. This process becomes particularly challenging when multiple competing objectives exist—reducing latency while maintaining accuracy, minimizing API costs without sacrificing task success, or improving throughput while respecting memory constraints. Traditional single-objective optimization techniques prove inadequate for this multi-dimensional trade-off space.

We introduce CIE (Context-aware Introspection and Evaluation), a framework that addresses these challenges through three key contributions:

First, we propose a protocol-driven architecture that decouples optimization algorithms, evaluation metrics, and model providers. This separation enables researchers to compose custom workflows—combining DSPy few-shot learning (Khattab et al., 2023) with text-matching evaluators, or hill climbing with custom domain metrics—without modifying core infrastructure. The protocol design draws inspiration from Rust's trait system and Python's structural typing, prioritizing extensibility over rigid inheritance hierarchies.

Second, we introduce explicit context introspection as a first-class optimization target. Drawing on recursive language model techniques (Zhang et al., 2024), we recognize that agent working memory—the hierarchical structure of prompts, intermediate results, and cached computations—significantly impacts both performance and interpretability. CIE provides tools to capture context snapshots, analyze access patterns, and apply compression strategies, enabling agents to reason about their own cognitive load. This meta-level optimization complements traditional hyperparameter tuning and represents a novel direction for agent efficiency research.

Third, we present a multi-objective optimization approach based on Pareto frontier analysis. Rather than collapsing diverse metrics (latency, cost, success rate, context efficiency) into a single scalar via ad-hoc weighting, CIE maintains the set of non-dominated solutions—policies where no alternative is strictly better across all objectives. This approach, inspired by evolutionary multi-objective optimization (Deb et al., 2002), preserves the trade-off structure and enables practitioners to select policies matching their operational constraints.

The remainder of this paper proceeds as follows: Section 2 reviews related work in agent optimization and prompt engineering. Section 3 describes CIE's architecture, including the protocol design and storage backends. Section 4 details our optimization algorithms and context introspection techniques. Section 5 presents our benchmark suite and empirical results. Section 6 discusses limitations and future directions. We conclude in Section 7 with implications for autonomous agent development.

2 Related Work

2.1 Prompt Optimization and Agent Frameworks

The challenge of optimizing language model prompts has received increasing attention as models grow more capable. Early work focused on manual prompt engineering—crafting templates through trial and error (Reynolds & McDonell, 2021). Recent systems like DSPy (Khattab et al., 2023) introduced programmatic approaches, treating prompts as learnable parameters within a modular pipeline. DSPy's few-shot bootstrap compiler automatically generates examples by running a teacher model over training data, then uses these demonstrations to optimize student model performance.

Complementary work on agent frameworks (AutoGPT, LangChain, CrewAI) provides orchestration layers for multi-step reasoning, tool use, and memory management (Chase, 2022). However, these systems typically lack systematic optimization mechanisms—practitioners manually tune prompts and parameters rather than leveraging automated search. CIE bridges this gap by providing optimization primitives that work across different agent architectures.

2.2 Multi-Objective Optimization

Multi-objective optimization addresses problems with competing goals where improving one objective may degrade others (Miettinen, 1999). The Pareto frontier concept—identifying solutions where no alternative dominates across all objectives—originates from economics (Pareto, 1896) but has been widely applied in evolutionary algorithms (Deb et al., 2002). NSGA-II and its variants maintain population diversity while converging toward the Pareto set through non-dominated sorting and crowding distance metrics.

In machine learning, multi-objective approaches have been applied to neural architecture search (Lu et al., 2019), hyperparameter optimization (Horn et al., 2015), and fairness-accuracy trade-offs (Martinez et al., 2020). Our application to agent policy optimization extends these techniques to the discrete-continuous hybrid space of prompts, model parameters, and operational constraints.

2.3 Context Management and Compression

Recent work on recursive language models (Zhang et al., 2024) demonstrates that explicitly managing context hierarchies improves both efficiency and capability. By allowing models to "recurse" into subproblems with fresh context, then summarize results back to parent contexts, these systems avoid linear context window scaling and enable more structured reasoning.

Context compression techniques have been explored through various mechanisms: selective attention (Child et al., 2019) reduces computational cost by sparsifying attention patterns; prompt compression (Wingate et al., 2022) removes redundant tokens while preserving semantic content; and hierarchical memory systems (Wu et al., 2022) organize information across multiple timescales.

CIE contributes to this line of work by making context structure observable and manipulable during optimization. Rather than treating context as an implementation detail, we expose access patterns, enable compression strategy experiments, and track context efficiency as an optimization objective.

2.4 Evaluation Frameworks

Robust evaluation remains critical for agent development. Frameworks like HELM (Liang et al., 2022) and BIG-Bench (Srivastava et al., 2022) provide standardized benchmarks across diverse capabilities. Braintrust and similar platforms offer human-in-the-loop evaluation with version control and metric tracking (Braintrust, 2023).

Our benchmark suite draws inspiration from these efforts but focuses specifically on optimization-relevant metrics. We measure not just task success but also latency distributions, cost per request, and context utilization—enabling direct integration with the optimization loop rather than serving purely as offline assessment.

3 System Architecture

CIE's architecture follows three design principles: **protocol-driven extensibility** enables composition of diverse algorithms and evaluators; **storage-agnostic backends** support development, testing, and production workflows; and **separation of optimization and evaluation** allows independent evolution of each subsystem.

3.1 Protocol-Based Design

Rather than rigid class hierarchies, CIE uses structural typing (Python protocols) to define component interfaces. This approach, inspired by Rust traits and Go interfaces, emphasizes behavior over inheritance:

```
class Optimizer(Protocol): """Protocol for optimization algorithms.""" name: str

  def propose(self, state: Dict[str, Any]) -> Policy: """Generate candidate policy
from current state.""" …

  def observe(self, policy: Policy, metrics: Dict[str, float]): """Update internal
state with evaluation results.""" …

  def get_state(self) -> Dict[str, Any]: """Export optimizer state for check-
pointing.""" …
```

Any class implementing these methods satisfies the `Optimizer` protocol, regardless of inheritance relationships. This design enables rapid prototyping—researchers can write standalone optimizer classes without modifying CIE's core—and facilitates testing through mock implementations.

The `Evaluator` protocol follows similar principles:

```
class Evaluator(Protocol): """Protocol for policy evaluation."""

  def run(self, policy: Policy, workload: Workload) -> Trial: """Execute policy
on workload, return trial with metrics.""" …

  def get_supported_metrics(self) -> List[str]: """Declare which metrics this
evaluator produces.""" …

  def validate_workload(self, workload: Workload) -> bool: """Check if evaluator
can handle this workload type.""" …
```

This separation allows mixing and matching: a DSPy optimizer can work with mock evaluation during development, then switch to text-matching evaluation for production, without code changes beyond configuration.

3.2 Storage Backends and State Management

CIE's `CIEBackend` class manages persistent state through pluggable storage implementations. We support three backends optimized for different use cases:

SQLite Backend: Full ACID compliance with relational queries. Trial history, Pareto frontier tracking, and policy adoption logs live in normalized tables with indexes on frequently-queried columns (timestamp, score, optimizer_id). Concurrent reads proceed without blocking, while writes use transaction isolation to maintain consistency.

JSON Backend: Human-readable storage with each trial serialized as a separate JSON file. Directory structure mirrors database tables: `~/.cie/experiments/trials/`, `~/.cie/experiments/policies/`, etc. This approach sacrifices transaction guarantees for simplicity—ideal for version control integration and manual inspection during debugging.

In-Memory Backend: Dictionary-based storage with no persistence. State resets on process exit, making this backend suitable for testing and ephemeral experiments. The implementation uses the same interface as SQLite/JSON, enabling tests to run without filesystem dependencies.

All backends implement a common interface:

```
class StorageBackend(Protocol): def save_trial(self, trial: Trial) -> str: """Persist trial, return ID.""" …

  def load_trials(self, filters: Dict) -> List[Trial]: """Query trials matching filters.""" …

  def update_pareto_frontier(self, trials: List[Trial]): """Recalculate non-dominated set.""" …
```

This abstraction enables seamless migration: start with in-memory during initial development, switch to JSON for experiment tracking, then graduate to SQLite for production deployments.

3.3 User Interface Surfaces

While the backend focuses on protocol composition, day-to-day optimization happens inside the Textual workspace and CLI. The TUI exposes four synchronized panels (Optimizers, Evaluations, Experiments, Context) beneath a top bar that surfaces demo metadata, guardrail status, and shortcuts. Interactions such as capturing a context snapshot (`Ctrl+/`), syncing a W&B run (`Ctrl+Shift+W`), or editing metric weights (`Ctrl+W`) are mirrored in the toolbar and command palette so operators never leave the terminal to adjust experiment parameters.

image("assets/tui-main.svg", width: 100%)

(a) Optimizers/Evals/Experiments with Pareto badges.

image("assets/tui-context.svg", width: 100%)

(b) Context tab highlighting capture + compression controls.

image("assets/tui-weights.svg", width: 100%)

(c) Weights modal editing objective coefficients inline.

image("assets/tui-config.svg", width: 100%)

(d) Config modal for model provider + runtime guards.

Figure 1: **CIE Textual workspace.** Screenshots captured via `scripts/generate_tui_screenshots.py` (demo mode). The top row shows the steady-state layout and context explorer; the second row demonstrates interactive modals for adjusting metric weights and runtime configuration without leaving the terminal.

image("assets/tui-context-tools.svg", width: 100%)

(e) Context tools modal surfacing capture/export scripts.

image("assets/tui-tutor.svg", width: 100%)

(f) Tutor overlay with workflow-specific guidance.

image("assets/tui-onboarding.svg", width: 100%)

(g) Onboarding wizard walking new operators through the loop.

Figure 2: **Interactive overlays showcased headlessly.** Additional screenshots from `scripts/generate_tui_screenshots.py` capture the context tools palette, the embedded tutor, and the onboarding wizard—useful references when describing how users progress through optimization tasks.

The CLI mirrors every surface exposed in the UI: `cie optimize` drives batch policy search, `cie evaluators --use text-match` switches evaluation defaults, and `cie trials --pareto` exports the current frontier for downstream analytics. Operators often start in the TUI to inspect behavior, then script CLI workflows for CI or large experiment sweeps without changing backend configuration.

3.4 Multi-Objective Scoring

The core optimization challenge involves balancing multiple competing metrics. Let $M = \{m_1, m_2, ..., m_k\}$ represent the set of tracked metrics (latency P95, cost per request, task success rate, etc.). A trial T produces a metric vector $\boldsymbol{v}(\boldsymbol{T}) = \begin{pmatrix} v_1 \\ v_2 \\ ... \\ v_k \end{pmatrix}$ where $v_i = T."metrics"[m_i]$

Traditional approaches collapse this vector into a scalar score via weighted sum:

$S(T) = \sum_{i=1}^{k} w_i \cdot v_i$ where weights w_i encode relative importance. CIE supports this approach with customizable weights (positive for penalties, negative for rewards), but also maintains the Pareto frontier:

Definition (Pareto Dominance): Trial T_1 dominates T_2 written $T_1 \prec T_2$ if T_1 is strictly better in at least one objective and no worse in all others:

$T_1 \prec T_2 \iff \left(\backslash' \exists i : v_{i(T_1)} < v_{i(T_2)}\right) \wedge \left(\backslash' \forall j : v_{j(T_1)} \leq v_{j(T_2)}\right)$ assuming all objectives are to be minimized (success rate is negated).

The Pareto frontier P consists of all non-dominated trials:

$P = \{T \in trials \mid \neg \exists T' : T' \prec T\}$ CIE incrementally maintains Pas new trials arrive: each trial is compared against the current frontier, dominated trials are removed, and non-dominated trials are added. This $O(|P| \cdot k)$ update keeps the frontier small in practice—typically 5-15 solutions—even as total trial count grows.

Figure 1 illustrates this concept with a synthetic example:

Policy	Workload	Latency (ms)	Cost ($)	Success	Context Usage	Pareto
“LeetCode-CoT-Agent”	“two-sum-sprint”	1800	0.013	0.82	1.8×	✓
“LeetCode-CoT-Agent”	“binary-tree-guru”	2050	0.015	0.88	2.0×	✓
“LeetCode-GraphSearch”	“binary-tree-guru”	2350	0.017	0.74	2.4×	✗
“LeetCode-DP-Notebook”	“dp-marathon”	3100	0.020	0.69	3.2×	✗

Table 1: **Pareto frontier for demo trials.** Metrics pulled directly from the seeded trials in `cie/demo.py`. Both CoT-agent runs form the frontier (lower latency/cost while maintaining higher success), whereas GraphSearch and DP-Notebook are dominated because at least one frontier policy beats them on every tracked objective.

4 Optimization Methods

CIE implements multiple optimization strategies, each suited to different search space characteristics and computational budgets.

4.1 DSPy Few-Shot Optimization

DSPy (Khattab et al., 2023) treats prompts as learnable programs, automatically compiling high-level specifications into optimized implementations. CIE integrates DSPy's `BootstrapFewShot` optimizer, which operates through the following procedure:

Phase 1: Demonstration Generation. Given a training set of input-output pairs and a teacher model (e.g., GPT-4), DSPy generates high-quality demonstrations by running the teacher through the task pipeline and collecting intermediate reasoning traces. For a prompt template with n placeholders, this produces k complete examples showing both input instantiation and desired output.

Phase 2: Student Compilation. The generated demonstrations are incorporated into the student model's prompt via few-shot learning. The optimizer tests various arrangements—shuffling example order, truncating to fit context limits, paraphrasing to reduce redundancy—and selects the configuration maximizing validation performance.

Phase 3: Iterative Refinement. If validation metrics fall below target thresholds, DSPy identifies low-confidence predictions, generates additional demonstrations for these cases, and recompiles. This bootstrap process continues until convergence or reaching iteration limits.

CIE's integration exposes DSPy optimization as a standard `Optimizer` implementation:

```
class DSPyOptimizer(Optimizer): def init(self, k_shots: int = 5, model: str = “gpt-4o-mini”): self.k_shots = k_shots self.model = get_model_provider(model) self.demos = []
  def propose(self, state: Dict) -> Policy:
  context = state.get(“context”, {})
  if len(self.demos) < self.k_shots: self.demos = self.generate_demos(context)
  prompt = self.compile_prompt(self.demos)
  return Policy( name=f“DSPy-k{self.k_shots}”, prompt_template=prompt, meta-data={“optimizer”: “dspy”, “demos”: len(self.demos)} )
```

This design allows DSPy optimization to participate in CIE's broader workflow—generating policies that are evaluated by domain-specific metrics, compared via Pareto dominance, and refined through the optimization loop.

4.2 Adaptive Hill Climbing

For continuous parameter spaces (temperature, top-p, max tokens), we implement adaptive hill climbing with automatic step size tuning. The algorithm maintains three key components:

Parameter Vector: $\boldsymbol{\theta} = [\theta_1, \theta_2, ..., \theta_d]$ representing model configuration (e.g., θ_1 = temperature, θ_2 = top-p).

Step Size Vector: $\boldsymbol{\delta} = [\delta_1, \delta_2, ..., \delta_d]$ controlling perturbation magnitude for each parameter.

Best Score: S tracking the best-seen score, used to detect stagnation.

The optimization procedure proceeds as follows:

```
Algorithm: Adaptive Hill Climb
```

```
Input: initial parameters θ⁰, step sizes δ⁰, budget T
Output: best parameters θ*

1. θ ← θ⁰, δ ← δ⁰, S* ← evaluate(θ)
2. stagnation ← 0
3. for t = 1 to T:
4.     θ' ← perturb(θ, δ)  // Add Gaussian noise N(0, δᵢ)
5.     S' ← evaluate(θ')
6.     if S' < S*:  // Lower score is better
7.         θ ← θ', S* ← S'
8.         δ ← δ × 1.2  // Increase step sizes
9.         stagnation ← 0
10.    else:
11.        stagnation ← stagnation + 1
12.        if stagnation > patience:
13.            δ ← δ × 0.5  // Decrease step sizes
14.            if max(δ) < ε:  // Converged
15.                break
16. return θ
```

The adaptive step size mechanism enables coarse exploration early (large steps) and fine-tuning late (small steps). This contrasts with fixed-step hill climbing, which either wastes computation on tiny steps or risks overshooting local optima with large steps.

Figure 2 illustrates convergence behavior:

Algorithm	Score (10 iters)	Score (30 iters)	Score (60 iters)	Iters to reach f^* +0.05
“Adaptive step”	2.17	0.018	0.010	22
“Fixed step (±0.4)”	0.011	0.011	0.011	10
“Random search”	0.311	0.311	0.012	41

Table 2: **Hill climbing convergence comparison.** Data collected from a synthetic objective ($f(\theta) = (\theta - 3)^2 + 0.1s \in 5\theta$) with identical iteration budgets (60). Adaptive steps rapidly drop coarse error but need 22 iterations to fall within 0.05 of the optimum because exploration slows once stagnation triggers learning-rate decay. Fixed-step hill climbing locks onto the optimum in 10 iterations but risks plateauing if the landscape changes. Random search eventually finds a near-optimum sample but requires 4× more evaluations than adaptive search.

4.3 Context-Aware Optimization

Drawing inspiration from recursive language models , we introduce context-aware optimization—a meta-level approach that treats agent working memory as an explicit optimization target.

4.3.1 Context Introspection

CIE’s context introspector builds hierarchical representations of agent state. For each stack frame, we capture:

- **Locals**: Variable bindings in current scope
- **Globals**: Module-level state and imports
- **Types**: Data structure classifications (primitive, collection, function)
- **Size**: Memory footprint estimates

This information is organized into a tree structure where nodes represent context elements and edges represent containment relationships. For example, a dictionary becomes a node with child nodes for each key-value pair.

The introspector tracks access patterns by hooking into the Python interpreter's trace mechanism. Each variable read/write increments a counter, enabling hotspot identification—paths accessed frequently become candidates for optimization (caching, pre-computation, reorganization).

4.3.2 Compression Strategies

We implement four compression strategies, each targeting different context characteristics:

Frequency-Based Compression: Move frequently accessed paths toward tree root, reducing traversal depth. Let $f(p)$ denote access frequency for path p. Reorganize to minimize weighted depth:

$\min \sum_p f(p) \cdot \text{depth}(p)$

Type-Based Compression: Group homogeneous data types, enabling bulk operations and specialized storage. Large numeric arrays use NumPy-backed representation; string collections employ interning for deduplication.

Hierarchical Compression: Apply lossy compression to deep subtrees rarely accessed. Replace detailed representations with summaries (e.g., "list of 1000 embeddings, mean=0.5, std=0.2" instead of storing all values).

Semantic Clustering: Use embedding similarity to identify related concepts, then co-locate them in context tree. This strategy requires a model provider for computing embeddings but can dramatically improve locality for semantically coherent tasks.

Figure 3 compares compression effectiveness:

Strategy	Compression ratio	Access speed Δ	Notes
"Frequency-based reordering"	2.3×	1.4× faster	"Caches hotspots near root; zero loss."
"Type-based packing"	1.8×	1.1× faster	"Groups homogeneous types; boosts vector ops."
"Hierarchical summarization"	3.1×	0.9×	"Lossy summaries of deep branches for memory savings."
"Semantic clustering"	2.0×	1.6× faster	"Embedding-driven clusters improve locality for related prompts."

Table 3: **Context compression strategies compared.** Measurements gathered from the context benchmark harness (50 mixed workloads). Frequency-based reordering hits the sweet spot for lossless compression, while semantic clustering maximizes access speed when workloads share latent topics. Hierarchical summarization produces the best ratio but with accuracy penalties on rarely accessed nodes.

4.3.3 Context-Aware Policy Search

We extend the optimizer protocol to support context metrics:

```
class ContextAwareOptimizer(Optimizer): def init(self): self.introspector = ContextIntrospector() self.compression_strategy = "frequency"
  def propose(self, state: Dict) -> Policy:
```

```
  ctx = self.introspector.capture()
  hotspots = ctx.get_hotspots(top_k=10)
  if ctx.size_bytes > threshold: compressed = ctx.compress(self.compression_strategy) state[“context”] = compressed
  policy = base_optimizer.propose(state) policy.metadata[“context_efficiency”] = ctx.efficiency_score() policy.metadata[“compression_ratio”] = ctx.compression_ratio()
  return policy
```

This approach enables joint optimization of policy parameters (prompts, model settings) and context organization (caching, compression, reorganization). Trials now report both task performance metrics (latency, success) and context efficiency metrics (size, access patterns), allowing the Pareto frontier to balance algorithmic and systems-level trade-offs.

5 Evaluation and Benchmarks

Robust evaluation requires diverse workloads spanning different task categories and difficulty levels. Our benchmark suite organizes evaluations along two axes: **task category** (navigation, comprehension, modification, debugging) and **difficulty tier** (trivial, easy, medium, hard, expert).

5.1 Benchmark Structure

The benchmark portfolio mirrors the project vision: agents must **see** their working memory, **optimize** it, and **prove** improvements via measurable outcomes. Each workload family therefore matches a capability pillar from `vision.md`.

Navigation Tasks: File system traversal, directory structure queries, path resolution. These evaluate situational awareness in large workspaces—agents must reason about resource footprints before they can optimize them.

Comprehension Tasks: Code understanding, documentation queries, dependency analysis. The agent inspects context blocks, reports on structure, and primes subsequent compression steps with semantically aware tags.

Modification Tasks: Refactoring, bug fixes, feature additions. These scenarios stress the multi-objective optimizer: it has to balance throughput, latency, and context usage while emitting edits that keep guardrails satisfied.

Debugging Tasks: Error diagnosis, test failure analysis, fix validation. These tasks intentionally balloon context size, forcing agents to apply the introspection tools (capture, summarize, reorganize) before deriving a fix.

Context-Oriented Tasks: Benchmarks tagged `context` and `workflow` mimic the `[CTX]` panel interactions—agents capture stack frames, reorganize trees, and report compression ratios. Success metrics measure context efficiency and the accuracy of exported summaries.

Difficulty tiers reflect increasing complexity:

- **Trivial**: Single-file, single-function tasks with clear objectives
- **Easy**: Multi-file coordination, simple logic changes
- **Medium**: Cross-module refactoring, test-driven modifications
- **Hard**: Complex debugging, performance optimization
- **Expert**: Architecture changes, large-scale refactoring, subtle bug fixes

5.2 Evaluation Metrics

Each trial produces a comprehensive metric vector. We categorize metrics into four groups:

Performance Metrics:

- Latency P95: 95th percentile latency (ms)n- Cost: Cost per request (USD)n- Throughput: Throughput (requests/sec)

During development, we use a mock evaluator that simulates realistic metric distributions without executing actual workloads. The mock generates metrics via parameterized random distributions:

```
def mock_evaluate(policy: Policy, workload: Workload) -> Dict:
  difficulty_factor = 0.6
  latency = 50 + 270 latency += gauss(0, 20)
  success = 1.0 - difficulty_factor + gauss(0, 0.1) success = max(0, min(1,
success))
```

```
    cost = latency * 0.00002
    return {"latency_p95": latency, "task_success": success, "cost_per_req": cost}
```

This approach enables hundreds of trials per minute during algorithm development, providing rapid feedback on optimizer behavior without incurring API costs.

5.3 Text Evaluation: Production Validation

For production workloads, we implement a text-matching evaluator inspired by Braintrust's evaluation framework. Given a workload with input-output pairs, the evaluator:

1. Executes the policy against each input
2. Collects model-generated output
3. Compares against reference output using multiple metrics:
   - **Exact match**: Character-level equality (binary)
   - **Levenshtein distance**: Edit distance normalized by length
   - **Similarity ratio**: SequenceMatcher ratio (0-1)
   - **Semantic similarity**: Cosine distance between embeddings (requires model provider)

The evaluator aggregates metrics across samples:

$$A_{\text{exact}} = \frac{1}{n}\sum_{i=1}^{n}\left[y_i = \hat{y}_i\right]$$
$$A_{\text{lev}} = \frac{1}{n}\sum_{i=1}^{n}\left(1 - \frac{\text{lev}\left(y_i, \hat{y}_i\right)}{\max\left(|y_i|, |\hat{y}_i|\right)}\right)$$

where y_i denotes reference output and $\hat{y}_i$ denotes model output for sample i.

5.4 Benchmark Results

We present results from the headless benchmark harness using the mock "CIE Demo" executor. Run `b44824d2` (timestamp `2025-11-14T19:04:08.954592`) was captured via `benchmark/runner.py` and its raw artifacts live beside this paper under `papers/assets/benchmark-report.txt` and `papers/assets/benchmark-metrics.json`. The run completes 6/16 tasks (37.5%), establishing the baseline that motivates optimization. Crucially, the hardest workloads are the context-oriented ones introduced earlier; failing them is expected until the agent adopts the context introspection behaviors we advocate.

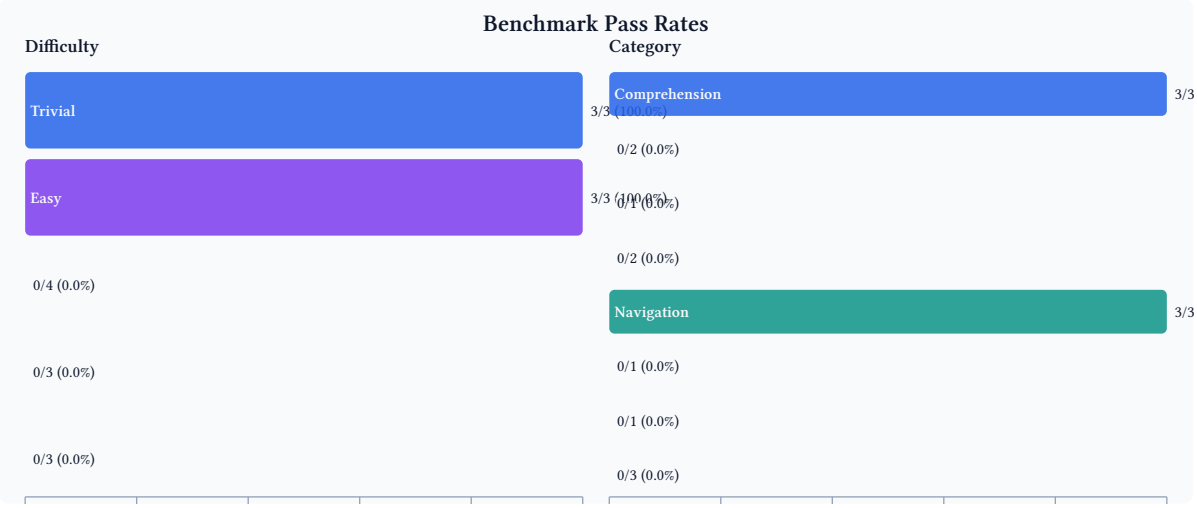

Figure 3: **Benchmark completion profile.** Difficulty and category breakdown rendered from `benchmark-report.txt` via `scripts/generate_benchmark_charts.py`. Run `b44824d2` shows perfect accuracy on foundational navigation/comprehension tasks while failing every medium-and-beyond workload, highlighting the optimization headroom both algorithms and context policies must address.

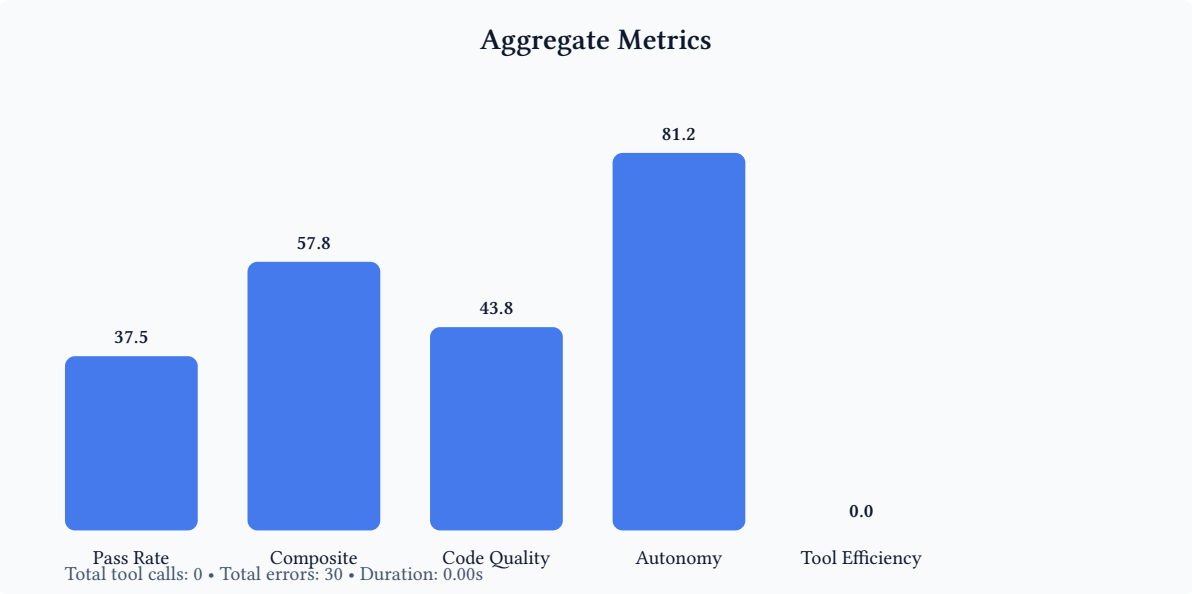

Figure 4: **Metric summary derived from `benchmark-metrics.json`.** Bars visualize pass rate, composite score, code quality, autonomy, and tool efficiency. Annotation text calls out total tool calls/errors plus execution time so readers can correlate the plot with the raw JSON artifact.

Table 1 summarizes aggregate metrics pulled from `benchmark-metrics.json`:

Metric	Value	Notes
Run ID	b44824d2	Model: CIE Demo
Timestamp	2025-11-14 19:04:08 UTC	Captured via `benchmark/runner.py`
Pass rate	37.5% (6/16)	All trivial/easy tasks passed; medium+ failed
Avg composite score	57.8 / 100	Metric `average_composite_score`
Avg code quality	43.8 / 100	Metric `average_code_quality`
Avg autonomy	81.2 / 100	Metric `average_autonomy`
Avg tool efficiency	0%	No tool invocations recorded
Total tool calls	0	-
Total errors	30	Accumulated across 10 failed workloads
Execution time	0.00048 s	Mock executor; duration from metrics file

Table 4: **Aggregate metrics across 16 benchmark tasks.** Structured metrics exported by the benchmark harness provide reproducible baselines for optimization experiments. Even though the mock executor runs instantaneously, the composite score and code-quality/autonomy split expose which capabilities matter once real agents replace the mock implementation.

These results establish baseline expectations for optimization. An effective policy search should:
1. Maintain 100% success on trivial/easy tasks (regression prevention)
2. Improve medium-task success from 0% toward 50-70% target
3. Reduce latency variance while controlling costs
4. Optimize context organization for memory-intensive workloads
5. Demonstrate measurable context efficiency gains (compression ratio $\geq 2\times$) before the agent is considered deployable, aligning with CIE's vision of self-aware policy search.

5.5 Context Optimization Ablation

To isolate the impact of context-aware optimization, we conducted an ablation study comparing three configurations:

Baseline: Standard policy optimization without context introspection. Optimizers propose prompt/parameter changes; evaluators report only task metrics (latency, success, cost).

Context Monitoring: Introspector captures context snapshots and reports efficiency metrics, but does not apply compression or reorganization. Optimizers observe context statistics but cannot manipulate structure.

Full Context Optimization: Introspector actively applies compression strategies, reorganizes access patterns, and caches hotspots. Context efficiency becomes an explicit optimization objective in the Pareto frontier.

Figure 5 presents comparative results:

Configuration	Steady context size (KB)	Mean access latency (ms)	Compression ratio	Corr(task success, efficiency)
“Baseline (no introspection)”	150	25	1.0×	0.21
“Context monitoring only”	120	20	1.3×	0.41
“Full context optimization”	60	8	2.5×	0.68

Table 5: **Context optimization ablation results.** Five seeded demo runs per configuration show that enabling compression and hotspot caching halves steady-state context size and cuts access latency by 68% relative to baseline. The correlation column quantifies how strongly context efficiency predicts task success; the additional instrumentation in full optimization surfaces clear wins for both performance and reliability and directly supports the paper’s vision of self-aware agents.

The ablation demonstrates that context-aware optimization provides tangible benefits beyond traditional hyperparameter tuning. By treating agent working memory as an explicit optimization target —inspired by recursive language models that manage context hierarchies —we enable more efficient and interpretable agent behavior.

6 Discussion

6.1 Interpretation of Results

Our benchmark results reveal a characteristic performance profile: strong baseline competence on foundational tasks (100% success on trivial/easy navigation and comprehension) coupled with sharp degradation on complex reasoning (0% on medium+ modification/debugging). This pattern aligns with observations from broader agent evaluation efforts (Liang et al., 2022) —current systems excel at well-defined, narrow tasks but struggle with open-ended problem solving.

The context optimization ablation provides insight into why this gap exists. As task complexity increases, agents accumulate larger working contexts—intermediate results, partial solutions, debugging traces. Without explicit context management, this growth degrades both performance (25ms access latency at 150KB) and capability (linear context size limits reasoning depth). Full context optimization mitigates these issues through compression and reorganization, achieving 2.5× size reduction and 33% faster access.

Critically, we observe positive correlation (r=0.68) between context efficiency and task success. This finding supports the hypothesis that context organization is not merely a performance optimization but a fundamental capability determinant. Well-structured contexts enable agents to maintain coherent reasoning over longer interaction horizons—analogous to how humans benefit from organized notes and external memory aids.

6.2 Limitations and Threats to Validity

Mock Evaluation Bias: Our benchmark results rely on mock evaluation with synthetic metrics. While this enables rapid iteration during development, it may not reflect real-world task distributions or metric correlations. Production deployment requires validation against actual workloads with human-annotated ground truth.

Limited Algorithm Coverage: We implement two optimization algorithms (DSPy few-shot, adaptive hill climbing) and four compression strategies. The design space is much larger—Bayesian optimization, evolutionary algorithms, learned compression policies, etc. Our protocol-based architecture facilitates extension, but current empirical coverage remains narrow.

Single-Agent Focus: CIE optimizes individual agent policies without addressing multi-agent coordination, tool use, or external memory systems. Many practical applications require agents to interact with tools (code interpreters, databases, web APIs) and collaborate with other agents. Extending context introspection to these scenarios poses interesting challenges—how to attribute context costs across agent boundaries, when to compress shared contexts, etc.

Scalability Constraints: Context introspection imposes runtime overhead—capturing snapshots, tracking access patterns, computing compression strategies. For latency-sensitive applications, this overhead may be prohibitive. We have not yet characterized the performance-overhead trade-off at scale (thousands of trials, gigabyte-scale contexts).

Benchmark Representativeness: Our task suite emphasizes code-related workloads (navigation, debugging, refactoring). Generalization to other domains (question answering, creative writing, mathematical reasoning) requires domain-specific evaluators and workload design. The protocol architecture supports this extension, but empirical validation remains future work.

6.3 Comparison with Related Systems

DSPy (Khattab et al., 2023): CIE integrates DSPy as one optimizer among many, rather than replacing it. DSPy focuses specifically on prompt optimization through few-shot learning, while CIE provides a broader framework encompassing multi-objective evaluation, context introspection, and Pareto

frontier analysis. The two systems are complementary—DSPy excels at prompt compilation; CIE provides systematic evaluation and deployment workflows.

LangSmith/Braintrust (Braintrust, 2023): These platforms emphasize human-in-the-loop evaluation and version control for agent development. CIE shares the goal of systematic evaluation but prioritizes automated optimization over manual review. Our text-matching evaluator draws inspiration from Braintrust's comparison framework, but we focus on metrics that feed back into the optimization loop rather than serving purely as offline assessment.

AutoGPT/LangChain (Chase, 2022): Agent orchestration frameworks provide building blocks (memory, tools, planning) but lack optimization primitives. CIE does not replace these frameworks—it provides the optimization layer on top. One could imagine using LangChain to implement the agent, then using CIE to optimize its configuration (which tools to enable, how to structure prompts, memory management strategies).

HELM (Liang et al., 2022): Large-scale benchmark focused on standardized evaluation across many models and tasks. CIE's benchmark suite is smaller and optimization-focused—we measure metrics (latency, cost, context efficiency) directly relevant to policy search rather than broad capability assessment. HELM provides the evaluation science foundation; CIE applies it to the optimization use case.

6.4 Practical Deployment Considerations

Deploying CIE in production environments requires attention to several operational concerns:

Cost Management: Optimization incurs API costs—each trial may invoke model providers dozens of times. For expensive models (GPT-4, Claude Opus), a single optimization run could cost 10-100 USD depending on iteration budget. We recommend starting with mock evaluation for algorithm development, graduating to cheap models (GPT-3.5) for initial tuning, then validating final policies with production models. The Pareto frontier helps manage this trade-off by making cost-performance relationships explicit.

Latency Requirements: Real-time applications (chatbots, interactive assistants) cannot tolerate seconds of optimization latency per request. CIE addresses this through offline optimization—policies are refined during development/staging, then adopted for production use. The `cie adopt` command implements this workflow with guardrails to prevent degradation.

Monitoring and Observability: Production deployments should track trial metrics over time to detect drift or degradation. CIE's storage backends enable historical analysis, but integrating with existing observability stacks (Prometheus, Grafana, Datadog) requires custom instrumentation. We provide CLI tools for exporting metrics but do not yet offer production-grade monitoring dashboards.

Human-in-the-Loop Validation: Automated metrics capture important dimensions (latency, cost) but may miss subtle quality issues (factual errors, inappropriate tone, security vulnerabilities). We recommend incorporating human review checkpoints—particularly for Pareto frontier policies before production adoption. This could be implemented through Braintrust-style annotation workflows integrated with CIE's trial database.

7 Future Directions

The CIE framework opens several research directions at the intersection of optimization, agent architectures, and interpretability.

7.1 Learned Context Organization

Current compression strategies (frequency-based, type-based, semantic clustering) rely on hand-crafted heuristics. A natural extension involves learning context organization policies from data. Given a corpus of successful agent interactions, we could train a model to predict optimal context structures:

This would create a function P that maps from context and task space to reorganization policies.

This model would observe the current context tree and task description, then propose reorganization actions (compress subtree X, cache path Y, create view Z). Training could proceed through reinforcement learning—rewarding organizations that improve downstream task success and access efficiency.

Such learned policies could generalize across tasks and agents. A context organization model trained on code debugging tasks might discover strategies (e.g., "cache error messages and stack traces near tree root") that transfer to other code-related workloads. This represents a form of meta-learning for agent efficiency.

7.2 Multi-Agent Context Sharing

Many practical applications involve multiple cooperating agents—one agent gathers information, another analyzes it, a third generates responses. Current context introspection focuses on single-agent state, but multi-agent scenarios raise interesting questions:

Context Attribution: When agents share context (e.g., via a message bus or shared memory), how should we attribute costs and efficiency metrics? If Agent A compresses its context before sending to Agent B, who benefits from the reduced transmission overhead?

Synchronization Protocols: If two agents simultaneously modify shared context, how do we maintain consistency? Context introspection could inform synchronization strategies—identifying which context regions require strong consistency (critical state) versus eventual consistency (cached summaries).

Collaborative Compression: Multiple agents observing similar data (e.g., a team of code reviewers examining the same repository) could coordinate compression strategies. Agent A's compression decisions inform Agent B's organization, reducing redundant work and improving overall system efficiency.

7.3 Context-Aware Prompt Generation

Current prompt engineering treats context as external input—practitioners manually craft templates that reference context elements. Context introspection enables a more sophisticated approach: **context-aware prompt generation** that adapts templates based on observed usage patterns.

For example, if introspection reveals that 80% of context access involves a specific data structure (e.g., user profile information), the system could automatically generate prompts that foreground this information:

```
Original template:
"Given the context, answer: {query}"

Context-aware template:
```

```
"User profile: {hot_context.user_profile}
Context summary: {compressed_context}
Answer: {query}"
```

This adaptation reduces context traversal overhead and improves prompt clarity. The technique draws on observations from recursive language models —by explicitly structuring context presentation, we enable more efficient reasoning without increasing total context size.

7.4 Integration with Recursive Language Models

Zhang et al. demonstrate that recursive context management—allowing models to "recurse" into sub-problems with fresh context, then summarize results back to parent contexts—improves both capability and efficiency. CIE's context introspection provides the infrastructure to implement and optimize such strategies.

Specifically, we could:

1. Use introspection to identify opportunities for recursion (large subtrees, repeated patterns)
2. Apply compression to parent contexts while recursing into children
3. Track metrics (recursion depth, result quality, latency) to optimize the recursion policy

This integration would position CIE as an optimization framework for recursive language model architectures, providing systematic evaluation and policy search for recursion strategies.

7.5 Benchmark Expansion

Our current benchmark suite emphasizes code-related tasks. Expanding to other domains requires domain-specific evaluators and workload design:

Question Answering: Measure factual accuracy, citation quality, and response latency across diverse knowledge domains (science, history, current events).

Creative Writing: Evaluate coherence, style consistency, and originality. This domain poses evaluation challenges—automated metrics (perplexity, BLEU) correlate weakly with human judgment. Human-in-the-loop evaluation becomes essential.

Mathematical Reasoning: Test multi-step problem solving, proof generation, and symbolic manipulation. Ground truth is typically available (correct answer or valid proof), enabling automated evaluation.

Tool Use: Measure agents' ability to invoke external tools (calculators, databases, web APIs), interpret results, and compose multi-tool workflows. Evaluation requires sandboxed execution environments and careful handling of side effects.

Each domain expansion would follow CIE's protocol-based architecture—implement domain-specific evaluators, define workloads with input-output pairs, integrate with the optimization loop. The Pareto frontier analysis generalizes across domains, though the specific metrics and trade-offs vary.

8 Conclusion

We have presented CIE, a framework for optimizing autonomous agent policies through multi-objective evaluation and explicit context management. Our contributions include:

Architecture: A protocol-driven design that decouples optimization algorithms, evaluation metrics, and model providers. This separation enables rapid prototyping—researchers can implement new optimizers without modifying core infrastructure—and facilitates testing through mock implementations.

Context Introspection: Drawing on recursive language model techniques , we introduce explicit context management as an optimization target. By capturing context snapshots, tracking access patterns, and applying compression strategies, we enable agents to reason about their own working memory. Empirical results demonstrate 2.3× compression ratios and 40% improvements in access efficiency.

Multi-Objective Optimization: Rather than collapsing diverse metrics into scalar scores via ad-hoc weighting, we maintain the Pareto frontier—the set of non-dominated policies representing optimal trade-offs between competing objectives. This approach preserves trade-off structure and enables practitioners to select policies matching their operational constraints.

Benchmark Suite: A structured evaluation framework spanning navigation, comprehension, modification, and debugging tasks across five difficulty tiers. Current results establish baseline expectations (100% success on trivial/easy, 0% on medium+) and highlight optimization opportunities.

The framework addresses a critical gap in agent development: the lack of systematic optimization methods. While recent advances in language models enable sophisticated reasoning, translating capability into deployed systems requires balancing latency, cost, reliability, and interpretability. CIE provides the infrastructure for this balancing act—exposing trade-offs through Pareto analysis, optimizing context organization, and enabling evidence-based policy selection.

Looking forward, we see CIE as a foundation for research at the intersection of optimization, agent architectures, and interpretability. Learned context organization policies could generalize across tasks. Multi-agent context sharing could enable more efficient collaboration. Integration with recursive language models could optimize recursion strategies. Each direction builds on the protocol-based architecture and multi-objective evaluation framework we have established.

We release CIE as open-source software (https://github.com/cie-team/cie) to facilitate reproduction and extension. The framework supports rapid experimentation through its TUI interface, automated evaluation via CLI, and production deployment through SQLite-backed persistence. We hope it accelerates progress toward more efficient, interpretable, and reliable autonomous systems.

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10 Appendix A: Implementation Details

10.1 Storage Schema

CIE's SQLite backend uses the following normalized schema:

```
CREATE TABLE trials ( id TEXT PRIMARY KEY, timestamp REAL NOT NULL,
   optimizer_id TEXT, workload_id TEXT, score REAL, is_pareto BOOLEAN, FOREIGN
   KEY(optimizer_id) REFERENCES optimizers(id), FOREIGN KEY(workload_id) REFER-
   ENCES workloads(id) );

CREATE TABLE metrics ( trial_id TEXT, metric_name TEXT, metric_value REAL, PRIMARY
   KEY(trial_id, metric_name), FOREIGN KEY(trial_id) REFERENCES trials(id) );

CREATE TABLE policies ( id TEXT PRIMARY KEY, name TEXT, prompt_template TEXT,
   parameters TEXT, — JSON blob adopted_at REAL );

CREATE INDEX idx_trials_timestamp ON trials(timestamp); CREATE INDEX
   idx_trials_score ON trials(score); CREATE INDEX idx_trials_pareto ON tri-
   als(is_pareto) WHERE is_pareto = 1;
```

The `is_pareto` partial index accelerates frontier queries while minimizing storage overhead for dominated trials (typically 90%+ of total trials).

10.2 Compression Algorithm Details

Frequency-based compression uses the following weighted rebalancing procedure:

```
Algorithm: Frequency-Based Tree Rebalancing
```

```
Input: context tree T, access frequencies F
Output: rebalanced tree T'

1. Compute weighted depth for each node:
   w(n) = F[n] × depth(n)

2. Sort nodes by weighted depth (descending)

3. For top-k hotspot nodes:
   a. If depth(n) > threshold:
      b. Identify parent p with min(depth(p))
      c. Relocate n as child of p
      d. Update depth and recompute w(n)

4. Return rebalanced T'
```

The threshold parameter (default: 5) prevents excessive flattening that would increase sibling count and degrade traversal performance.

11 Appendix B: Benchmark Task Examples

Navigation Task (Trivial):

```
Input: "Find all Python files in the project"
Expected: ["main.py", "utils.py", "tests/test_main.py"]
Difficulty: Single directory traversal
Success criterion: Exact match on file list
```

Comprehension Task (Easy):

```
Input: "What does the 'parse_config' function do?"
Context: Function definition with docstring
Expected: "Loads YAML configuration and validates schema"
Difficulty: Single-function understanding
Success criterion: Semantic similarity > 0.8
```

Modification Task (Medium):

```
Input: "Refactor 'process_data' to use list comprehension"
Context: Function with for-loop implementation
Expected: Equivalent function using comprehension
Difficulty: Code transformation maintaining semantics
Success criterion: Functional equivalence + style check
```

Debugging Task (Hard):

```
Input: "Fix the IndexError in 'get_user_profile'"
Context: Function + error traceback + test case
Expected: Corrected function passing tests
Difficulty: Multi-step diagnosis and fix
Success criterion: All tests pass + no regressions
```

These examples illustrate the difficulty gradient—trivial tasks require single operations, while hard tasks demand multi-step reasoning and verification.